

```

1  -- Code your testbench here
2  library IEEE;
3  use IEEE.std_logic_1164.all;
4  use IEEE.numeric_std.all;
5
6  entity test1 is end test1;
7
8  architecture test of test1 is
9      -->signal x : unsigned(7 downto 0);
10     -->signal y : unsigned(2 downto 0);
11
12     begin
13         -->testEnc: entity work.enc8_3
14         -->port map ( x=>x, y=>y );
15         -->process
16         -->variable testInput : unsigned(7 downto 0);
17         -->begin
18         -->testInput:=to_unsigned(1,8);
19         -->for i in 0 to 7 loop
20         -->-->x<=testInput;
21         -->-->wait for 10 ns;
22         -->-->report "x = " & integer'image(to_integer(x)) & " y = " &
23         -->-->integer'image(to_integer(y));
24         -->-->testInput:= shift_left(testInput,1);
25         -->-->end loop;
26         -->-->wait;
27         -->end process;
28     end test;
29
30     -- Code your design here
31     library IEEE;
32     use IEEE.std_logic_1164.all;
33     use IEEE.numeric_std.all;
34
35     entity enc8_3 is
36         -->port( x : in unsigned(7 downto 0);
37         -->-->y : out unsigned(2 downto 0)
38         -->-->);
39     end enc8_3;
40
41     architecture core_enc of enc8_3 is
42     begin
43         -->process(x)
44         -->begin
45         -->-->case to_integer(x) is
46         -->-->when 1 => y <= to_unsigned(0,3);
47         -->-->when 2 => y <= to_unsigned(1,3);
48         -->-->when 4 => y <= to_unsigned(2,3);
49         -->-->when 8 => y <= to_unsigned(3,3);
50         -->-->when 16 => y <= to_unsigned(4,3);
51         -->-->when 32 => y <= to_unsigned(5,3);
52         -->-->when 64 => y <= to_unsigned(6,3);
53         -->-->when 128 => y <= to_unsigned(7,3);
54         -->-->when others => y<="000";
55         -->-->end case;
56         -->end process;
57     end core_enc;
58
59     [2026-04-23 23:06:44 UTC] ghdl -i design.vhd testbench.vhd && ghdl -m test1 && ghdl -r
60     test1
61     analyze design.vhd
62     analyze testbench.vhd
63     elaborate test1
64     ../../src/ieee/v93/numeric_std-body.vhdl:2098:7:@0ms:(assertion warning):
65     NUMERIC_STD.TO_INTEGER: metavalue detected, returning 0
66     testbench.vhd:22:13:@10ns:(report note): x = 1 y = 0
67     testbench.vhd:22:13:@20ns:(report note): x = 2 y = 1

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63 testbench.vhd:22:13:@30ns:(report note): x = 4 y = 2
64 testbench.vhd:22:13:@40ns:(report note): x = 8 y = 3
65 testbench.vhd:22:13:@50ns:(report note): x = 16 y = 4
66 testbench.vhd:22:13:@60ns:(report note): x = 32 y = 5
67 testbench.vhd:22:13:@70ns:(report note): x = 64 y = 6
68 testbench.vhd:22:13:@80ns:(report note): x = 128 y = 7
69
70
71
72
73
74
75
76 -- Code your design here
77 library IEEE;
78 use IEEE.std_logic_1164.all;
79
80 entity mux_16_1 is
81     port(
82         Xin : in STD_LOGIC_VECTOR(15 downto 0);
83         n : in integer range 0 to 15;
84         Yout : out STD_LOGIC );
85 end mux_16_1;
86
87 architecture mux_core of mux_16_1 is
88 begin
89     process( Xin, n )
90     begin
91         Yout<=Xin(n);
92     end process;
93 end mux_core;
94
95
96 -- Code your testbench here
97 library IEEE;
98 use IEEE.std_logic_1164.all;
99 use IEEE.numeric_std.all;
100
101 entity test2 is end test2;
102
103 architecture tester of test2 is
104     signal Xin : STD_LOGIC_VECTOR(15 downto 0);
105     signal n : integer range 0 to 15;
106     signal Yout : STD_LOGIC;
107 begin
108     TestItem : entity work.mux_16_1
109     port map (Xin=>Xin, n=>n, Yout=>Yout);
110     process
111     begin
112         Xin<="0101010101010101";
113         for i in 0 to 15 loop
114             n<=i;
115             wait for 10 ns;
116             report "n = " & integer'image(n) & " Yout = " & std_logic'image(Yout);
117         end loop;
118         wait;
119     end process;
120 end tester;
121
122 [2026-04-23 23:25:46 UTC] ghdl -i design.vhd testbench.vhd && ghdl -m test2 && ghdl -r
test2
123 analyze design.vhd
124 analyze testbench.vhd
125 elaborate test2
126 testbench.vhd:21:13:@10ns:(report note): n = 0 Yout = '1'

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127 testbench.vhd:21:13:@20ns:(report note): n = 1 Yout = '0'
128 testbench.vhd:21:13:@30ns:(report note): n = 2 Yout = '1'
129 testbench.vhd:21:13:@40ns:(report note): n = 3 Yout = '0'
130 testbench.vhd:21:13:@50ns:(report note): n = 4 Yout = '1'
131 testbench.vhd:21:13:@60ns:(report note): n = 5 Yout = '0'
132 testbench.vhd:21:13:@70ns:(report note): n = 6 Yout = '1'
133 testbench.vhd:21:13:@80ns:(report note): n = 7 Yout = '0'
134 testbench.vhd:21:13:@90ns:(report note): n = 8 Yout = '1'
135 testbench.vhd:21:13:@100ns:(report note): n = 9 Yout = '0'
136 testbench.vhd:21:13:@110ns:(report note): n = 10 Yout = '1'
137 testbench.vhd:21:13:@120ns:(report note): n = 11 Yout = '0'
138 testbench.vhd:21:13:@130ns:(report note): n = 12 Yout = '1'
139 testbench.vhd:21:13:@140ns:(report note): n = 13 Yout = '0'
140 testbench.vhd:21:13:@150ns:(report note): n = 14 Yout = '1'
141 testbench.vhd:21:13:@160ns:(report note): n = 15 Yout = '0'
142
143
144
145
146
147
148
149 -- Code your design here
150 library IEEE;
151 use IEEE.std_logic_1164.all;
152
153
154 entity shift_reg_4 is
155     —>port(
156         clk: in STD_LOGIC;
157         rst: in STD_LOGIC;
158         en: in STD_LOGIC;
159         x: in STD_LOGIC;
160         Yout : out STD_LOGIC_VECTOR(3 downto 0));
161 end shift_reg_4;
162
163 architecture sr4_core of shift_reg_4 is
164     —>signal int_y : STD_LOGIC_VECTOR( 3 downto 0) := "0000";
165 begin
166     —>process(clk, rst)
167     —>begin
168         —>if(rst='1') then
169             —>int_y<="0000";
170         —>elsif rising_edge(clk) and en='1' then
171             —>int_y<=x & int_y(3 downto 1);
172         —>end if;
173     —>end process;
174     —>Yout<=int_y;
175 end sr4_core;
176
177
178
179 -- Code your testbench here
180 library IEEE;
181 use IEEE.std_logic_1164.all;
182
183 entity test3 is end test3;
184
185 architecture tester of test3 is
186     —>signal clk: STD_LOGIC:='0';
187     —>signal rst: STD_LOGIC:='0';
188     —>signal en: STD_LOGIC:='0';
189     —>signal x: STD_LOGIC:='0';
190     —>signal Yout: STD_LOGIC_VECTOR(3 downto 0);
191     —>constant clk_period : time := 10 ns;

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192 begin
193   →testItem : entity work.shift_reg_4
194     →port map(
195       →clk=>clk,
196       →rst=>rst,
197       →en=>en,
198       →x=>x,
199       →Yout=>Yout);
200   →clk<= not clk after clk_period/2;
201   process
202   begin
203     →rst<='1';
204     en<='0';
205     x<='0';
206     wait for 15 ns;
207     rst<='0';
208     en<='1';
209     wait for clk_period;
210
211     report "Yout = " & to_string(Yout);
212     x<='1'; wait for clk_period;
213     report "Yout = " & to_string(Yout);
214     x<='0'; wait for clk_period;
215     report "Yout = " & to_string(Yout);
216     x<='1'; wait for clk_period;
217     report "Yout = " & to_string(Yout);
218     x<='0'; wait for clk_period;
219     report "Yout = " & to_string(Yout);
220     x<='1'; wait for clk_period;
221     report "Yout = " & to_string(Yout);
222     x<='0'; wait for clk_period;
223     report "Yout = " & to_string(Yout);
224     x<='1'; wait for clk_period;
225     report "Yout = " & to_string(Yout);
226     report "TEST FINISHED" severity failure;
227   end process;
228 end tester;
229
230
231 [2026-04-23 23:59:35 UTC] ghdl -i '--std=08' design.vhd testbench.vhd && ghdl -m '--std=08'
test3 && ghdl -r test3
232 analyze design.vhd
233 analyze testbench.vhd
234 elaborate test3
235 testbench.vhd:33:9:@25ns:(report note): Yout = 0000
236 testbench.vhd:35:9:@35ns:(report note): Yout = 0000
237 testbench.vhd:37:9:@45ns:(report note): Yout = 1000
238 testbench.vhd:39:9:@55ns:(report note): Yout = 0100
239 testbench.vhd:41:9:@65ns:(report note): Yout = 1010
240 testbench.vhd:43:9:@75ns:(report note): Yout = 0101
241 testbench.vhd:45:9:@85ns:(report note): Yout = 1010
242 testbench.vhd:47:9:@95ns:(report note): Yout = 0101
243 testbench.vhd:48:9:@95ns:(report failure): TEST FINISHED
244 ./test3:error: report failed
245   instance: .test3(test3).P1
246 ./test3:error: simulation failed
247 Exit code expected: 0, received: 1

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